Dashboard Query's

KPI's

1. Total loan application:
select count(id) as Total_Applications from financial_loan_table

Total_Applications
38576

2. MTD loan application:
Select count(id) as MTD_Total_Applications from financial_loan_table

Where MONTH(issue_date) = 12

MTD_Total_Applications
4314

3. PMTD loan application:
select count(id) as PMTD_Total_Applications from financial_loan_table
WHERE issue_date >= '2021-11-01' AND issue_date < '2021-12-31';

PMTD_Total_Applications
8349

==========

4. Total funded amount:
select format(SUM(loan_amount), 'C0') as Total_funded_amount
from financial_loan_table;

Total_funded_amount
$435,757,075

5. MTD Total funded amount:
SELECT FORMAT(SUM(loan_amount), 'C0') AS MTD_Total_Funded_Amount
FROM financial_loan_table
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;

MTD_Total_Funded_Amount
$53,981,425

6.PMTD Total funded amount:
SELECT FORMAT(SUM(loan_amount), 'C0') AS PMTD_Total_Funded_Amount
FROM financial_loan_table
WHERE issue_date >= '2021-11-01' AND issue_date < '2021-12-31';

PMTD_Total_Funded_Amount
$101,736,250

SHREYASH_ANALYSIS_QUERIES

7. Total Received Amount:

```
SELECT Format(SUM(total_payment), 'c0') as Total_Received_Amount from financial_loan_table;
```

Total_Received_Amount
$473,070,933

8. MTD Total Amount Received:

```
SELECT Format(SUM(total_payment), 'c0') as Total_Received_Amount from financial_loan_table
where MONTH(issue_date) = 12;
```

MTD_Total_Received_Amount
$58,074,380

9. MTD Total Amount Received:

```
SELECT Format(SUM(total_payment), 'c0') as PMTD_Total_Received_Amount from financial_loan_table
WHERE issue_date >= '2021-11-01' AND issue_date <  '2021-12-31';
```

PMTD_Total_Received_Amount
$108,206,410

==========

10. Average Interest Rate:

```
select format(AVG(int_rate), 'p2') as AVG_Interest_Rate from financial_loan_table;
```

AVG_Interest_Rate
12.05%

11.Month over Month change:

```
WITH monthly AS (
SELECT DATEFROMPARTS(YEAR(issue_date), MONTH(issue_date), 1) AS month_start,
AVG(int_rate) AS avg_int_rate FROM financial_loan_table
GROUP BY DATEFROMPARTS(YEAR(issue_date), MONTH(issue_date), 1) ),
with_prev AS (
SELECT month_start, avg_int_rate,
LAG(avg_int_rate) OVER (ORDER BY month_start) AS prev_avg_int_rate
FROM monthly )
SELECT
 FORMAT(month_start, 'yyyy-MM') AS month,
 FORMAT(avg_int_rate, 'P2') AS avg_interest_rate,
 FORMAT(prev_avg_int_rate, 'P2') AS prev_month_rate,
 FORMAT((avg_int_rate - prev_avg_int_rate) / NULLIF(prev_avg_int_rate, 0), 'P2') AS mom_change
 FROM with_prev
ORDER BY month_start;
```

SHRIYASH_ANALYSIS_QUERIES

	month	avg_interest_rate	prev_month_rate	mom_change
1	2021-01	11.46%	NULL	NULL
2	2021-02	11.72%	11.46%	2.27%
3	2021-03	11.86%	11.72%	1.17%
4	2021-04	11.74%	11.86%	-0.99%
5	2021-05	12.26%	11.74%	4.40%
6	2021-06	12.27%	12.26%	0.13%
7	2021-07	12.24%	12.27%	-0.30%
8	2021-08	12.30%	12.24%	0.51%
9	2021-09	12.00%	12.30%	-2.41%
10	2021-10	12.02%	12.00%	0.17%
11	2021-11	11.94%	12.02%	-0.69%
12	2021-12	12.36%	11.94%	3.47%

12.Average Debt-to-income-ratio (DTI):

```
select FORMAT(avg(dti), 'p2') as AVG_DTI from financial_loan_table;
```

	AVG_DTI
1	13.33%

13.MTD AVG_DTI:

```
SELECT FORMAT(AVG(dti), 'P2') AS mtd_avg_dti
FROM financial_loan_table
WHERE issue_date >= (SELECT DATEFROMPARTS(YEAR(MAX(issue_date)), MONTH(MAX(issue_date)), 1)
FROM financial_loan_table);
```

	mtd_avg_dti
1	13.67%

14.MOM CHANGE IN DTI:

```
WITH monthly AS
(SELECT DATEFROMPARTS(YEAR(issue_date), MONTH(issue_date), 1) AS month_start, AVG(dti) AS avg_dti
FROM financial_loan_table
GROUP BY DATEFROMPARTS(YEAR(issue_date), MONTH(issue_date), 1)),
with_prev AS (SELECT month_start, avg_dti,
LAG(avg_dti) OVER (ORDER BY month_start) AS prev_avg_dti
FROM monthly)
SELECT
FORMAT(month_start, 'yyyy-MM') AS month,
FORMAT(avg_dti, 'P2') AS avg_dti,
FORMAT(prev_avg_dti, 'P2') AS prev_avg_dti,
FORMAT((avg_dti - prev_avg_dti) / NULLIF(prev_avg_dti, 0), 'P2') AS mom_change
FROM with_prev
ORDER BY month_start;
```

SHREYASH_ANALYSIS_QUERIES

	month	avg_dti	prev_avg_dti	mom_change
1	2021-01	12.94%	NULL	NULL
2	2021-02	13.41%	12.94%	3.65%
3	2021-03	13.22%	13.41%	-1.44%
4	2021-04	13.22%	13.22%	0.03%
5	2021-05	13.33%	13.22%	0.87%
6	2021-06	13.24%	13.33%	-0.67%
7	2021-07	13.29%	13.24%	0.39%
8	2021-08	13.35%	13.29%	0.44%
9	2021-09	13.30%	13.35%	-0.41%
10	2021-10	13.41%	13.30%	0.88%
11	2021-11	13.30%	13.41%	-0.83%
12	2021-12	13.67%	13.30%	2.73%

15.GOOD LOAN PCT:

```
SELECT
FORMAT( CAST(COUNT(CASE WHEN LOAN_STATUS IN ('FULLY PAID', 'CURRENT') THEN ID END) AS DECIMAL(10,4))
/ COUNT(ID),
'P2' ) AS GOOD_LOAN_PCT
FROM financial_loan_table;
```

	GOOD_LOAN_PCT
1	86.18%

16.GOOD LOAN APPLICATIONS:

```
SELECT COUNT(ID) AS GOOD_LOAN_APPLICATION FROM financial_loan_table
WHERE loan_status = 'FULLY PAID' OR loan_status = 'CURRENT';
```

	GOOD_LOAN_APPLICATION
1	33243

17.GOOD LOAN AMOUNT FUNDED:

```
SELECT FORMAT(SUM(loan_amount),'N2') AS GOOD_LOAN_FUNDED FROM financial_loan_table
WHERE loan_status = 'FULLY PAID' OR loan_status = 'CURRENT';
```

	GOOD_LOAN_FUNDED
1	370,224,850.00

18. GOOD_LOAN_RECEIVED:

```
SELECT FORMAT(SUM(total_payment),'N2') AS GOOD_LOAN_RECEIVED FROM financial_loan_table
WHERE loan_status IN ('FULLY PAID','CURRENT');
```

	GOOD_LOAN_RECEIVED
1	435,786,170.00

19.BAD LOAN PCT:

```
SELECT
FORMAT( CAST(COUNT(CASE WHEN LOAN_STATUS IN ('CHARGED OFF') THEN ID END) AS DECIMAL(10,4))
/ COUNT(ID),
'P2' ) AS BAD_LOAN_PCT
```

SURYAKASH_ANALYSIS_QUERIES

FROM financial_loan_table;

	BAD_LOAN_PCT
1	13.82%

20.BAD LOAN APPLICATIONS:

SELECT COUNT(ID) AS BAD_LOAN_APPLICATION FROM financial_loan_table
WHERE loan_status = 'CHARGED OFF';

	BAD_LOAN_APPLICATION
1	5333

21.BAD LOAN AMOUNT FUNDED:

SELECT FORMAT(SUM(loan_amount),'N2') AS BAD_LOAN_FUNDED FROM financial_loan_table
WHERE loan_status = 'CHARGED OFF';

	BAD_LOAN_FUNDED
1	65,532,225.00

23. BAD_LOAN_RECEIVED:

SELECT FORMAT(SUM(total_payment),'N2') BAD_LOAN_RECEIVED FROM financial_loan_table
WHERE loan_status IN ('CHARGED OFF');

	BAD_LOAN_RECEIVED
1	37,284,763.00

24. LOAN STATUS:

```
SELECT
LOAN_STATUS,
COUNT(ID) AS TOTAL_APPLICATION,
SUM(LOAN_AMOUNT) AS TOTAL_AMONT_FUNDED,
SUM(TOTAL_PAYMENT) AS TOTAL_AMOUNT_RECEIVED,
FORMAT(AVG(DTI),'P2') AS AVG_DTI,
FORMAT(AVG(INT_RATE),'P2') AS AVG_INT
FROM financial_loan_table
GROUP BY LOAN_STATUS;
```

	LOAN_STATUS	TOTAL_APPLICATION	TOTAL_AMONT_FUNDED	TOTAL_AMOUNT_RECEIVED	AVG_DTI	AVG_INT
1	Fully Paid	32145	351358350	411586256	13.17%	11.64%
2	Charged Off	5333	65532225	37284763	14.00%	13.88%
3	Current	1098	18866500	24199914	14.72%	15.10%

25.LOAN STATUS MTD:

```
SELECT
LOAN_STATUS,
FORMAT(SUM(LOAN_AMOUNT), 'N0') AS MTD_LOAN_AMOUNT,
FORMAT(SUM(TOTAL_PAYMENT), 'N0') AS MTD_TOTAL_PAYMENT
FROM financial_loan_table
WHERE ISSUE_DATE >= (SELECT DATEFROMPARTS(YEAR(MAX(ISSUE_DATE)), MONTH(MAX(ISSUE_DATE)), 1)
FROM financial_loan_table)
```

SHRAVASTI_ANALYSIS_QUERIES

GROUP BY LOAN_STATUS;

	LOAN_STATUS	MTD_LOAN_AMOUNT	MTD_TOTAL_PAYMENT
1	Fully Paid	41,302,025	47,815,851
2	Charged Off	8,732,775	5,324,211
3	Current	3,946,625	4,934,318

CHART 2:

1.MONTHLY TREND:

```
SELECT
   DATENAME(MONTH, issue_date) AS Month_Name,
   COUNT(id) AS Total_Loan_Applications,
   format(SUM(loan_amount),'N0') AS Total_Funded_Amount,
   FORMAT(SUM(total_payment),'N0') AS Total_Received_Amount
FROM financial_loan_table
GROUP BY DATENAME(MONTH, issue_date), MONTH(issue_date)
ORDER BY MONTH(issue_date);
```

	Month_Name	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	January	2332	25,031,650	27,578,836
2	February	2279	24,647,825	27,717,745
3	March	2627	28,875,700	32,264,400
4	April	2755	29,800,800	32,495,533
5	May	2911	31,738,350	33,750,523
6	June	3184	34,161,475	36,164,533
7	July	3366	35,813,900	38,827,220
8	August	3441	38,149,600	42,682,218
9	September	3536	40,907,725	43,983,948
10	October	3796	44,893,800	49,399,567
11	November	4035	47,754,825	50,132,030
12	December	4314	53,981,425	58,074,380

2.REGIONAL ANALYSIS BY STATE:

```
SELECT
   ADDRESS_STATE,
   COUNT(id) AS Total_Loan_Applications,
   format(SUM(loan_amount),'N0') AS Total_Funded_Amount,
   FORMAT(SUM(total_payment),'N0') AS Total_Received_Amount
FROM financial_loan_table
GROUP BY ADDRESS_STATE
ORDER BY COUNT(id) DESC;
```

3.LOAN TERM ANALYSIS:

```
SELECT
   TERM,
   COUNT(id) AS Total_Loan_Applications,
   format(SUM(loan_amount),'N0') AS Total_Funded_Amount,
   FORMAT(SUM(total_payment),'N0') AS Total_Received_Amount
FROM financial_loan_table
GROUP BY TERM
ORDER BY COUNT(id) DESC;
```

SHREYASH_A_QUERIES

	TERM	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	36 months	28237	273,041,225	294,709,458
2	60 months	10339	162,715,850	178,361,475

4.EMPLYOEE LENGTH:

```
SELECT
   emp_length,
   COUNT(id) AS Total_Loan_Applications,
   format(SUM(loan_amount),'N0') AS Total_Funded_Amount,
   FORMAT(SUM(total_payment),'N0') AS Total_Received_Amount
FROM financial_loan_table
GROUP BY emp_length
ORDER BY COUNT(id) DESC;
```

	emp_length	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	10+ years	8870	116,115,950	125,871,616
2	< 1 year	4575	44,210,625	47,545,011
3	2 years	4382	44,967,975	49,206,961
4	3 years	4088	43,937,850	47,551,832
5	4 years	3428	37,600,375	40,964,850
6	5 years	3273	36,973,625	40,397,571
7	1 year	3229	32,883,125	35,498,348
8	6 years	2228	25,612,650	27,908,658
9	7 years	1772	20,811,725	22,584,136
10	8 years	1476	17,558,950	19,025,777
11	9 years	1255	15,084,225	16,516,173

5.PURPOSE:

```
SELECT
   purpose,
   COUNT(id) AS Total_Loan_Applications,
   format(SUM(loan_amount),'N0') AS Total_Funded_Amount,
   FORMAT(SUM(total_payment),'N0') AS Total_Received_Amount
FROM financial_loan_table
GROUP BY purpose
ORDER BY COUNT(id) DESC;
```

	purpose	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	Debt consolidation	18214	232,459,675	253,801,871
2	credit card	4998	58,885,175	65,214,084
3	other	3824	31,155,750	33,289,676
4	home improvement	2876	33,350,775	36,380,930
5	major purchase	2110	17,251,600	18,676,927
6	small business	1776	24,123,100	23,814,817
7	car	1497	10,223,575	11,324,914
8	wedding	928	9,225,800	10,266,856
9	medical	667	5,533,225	5,851,372
10	moving	559	3,748,125	3,999,899
11	house	366	4,824,925	5,185,538
12	vacation	352	1,967,950	2,116,738
13	educational	315	2,161,650	2,248,380
14	renewable_energy	94	845,750	898,931

6.HOME OWNERSHIP:

```
SELECT
   home_ownership,
   COUNT(id) AS Total_Loan_Applications,
```

```
    format(SUM(loan_amount),'N0') AS Total_Funded_Amount,
    FORMAT(SUM(total_payment),'N0') AS Total_Received_Amount
FROM financial_loan_table
GROUP BY home_ownership
ORDER BY COUNT(id) DESC;
```

	home_ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	RENT	18439	185,768,475	201,823,056
2	MORTGAGE	17198	219,329,150	238,474,438
3	OWN	2838	29,597,675	31,729,129
4	OTHER	98	1,044,975	1,025,257
5	NONE	3	16,800	19,053

SHREYASH_ANALYSIS_QUERIES